

Validation of an accessible tool for component-based multidisciplinary analysis of electric fixed-wing UAVs

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State-of-the-art, open-source multidisciplinary analysis codes applicable to small electric fixed-wing UAVs are generally divorced from popular tools used for component-based propulsion selection. This paper introduces UAVDEX, an accessible Python-based tool with integrated commercial off-the-shelf (COTS) component databases and electric-propulsion modeling. The code bridges the gap between existing tools that provide gradient-based optimization of electric UAVs, and common hobbyist calculators used to evaluate COTS propulsion drives. This paper describes UAVDEX's core propulsion model and related validation efforts. The results suggest that the model provides adequate estimation of relevant design parameters for early-stage UAV design, especially for runtime prediction. Case studies demonstrate the code's use for design and analysis of the Rensselaer Polytechnic Institute's 2026 Design/Build/Fly (DBF) aircraft.

I. Introduction

While most full-scale aircraft are limited to a handful of engine options, the electric unmanned aerial vehicle (UAV) designer is presented with a vast array of commercial off-the-shelf (COTS) batteries, motors, and propellers. This work aims to supply UAV designers, specifically student design teams, with tools to accelerate the design process when numerous COTS components are available to select from.

Key performance parameters for small fixed-wing aircraft used in competitions such as AIAA Design/Build/Fly (DBF) or SAE Aero Design include cruise/level turn velocity, range (R), endurance (t), and specific excess power (P_s). A prerequisite to predicting steady-state velocity and P_s is accurately determining thrust (T) with respect to velocity (U_∞), throttle setting (δT), air density (ρ), and battery cell voltage (V_{oc}). Critically, electric aircraft differ from conventional aircraft in that they will have significantly decreased thrust as the state of charge (SOC) of the battery decreases [1]. Thus, the cruise and level turn velocities of an all-electric aircraft will substantially decrease throughout a given mission. In the 2025 DBF fly-off, this decrease in flight speed caused vicious stalls that were responsible for the majority of crashes.

What tools are publicly available for the student designer or hobbyist looking to select their propulsion? There exists an enormous catalog of calculators created for small-scale electric remote-control (RC) aircraft[‡], plus several programs of academic origin. These calculators are somewhat accurate for static thrust and electrical quantities, such as current, voltage, and power at fixed conditions. Some provide models for propeller thrust as it varies over different freestream velocities (U_∞) as shown in Figure 1. However, these accessible calculators typically assume a fixed battery cell voltage, and, moreover, they provide poor runtime predictions, largely due to inaccurate propeller modeling. Regarding academic tools, Mark Drela's popular QPROP code enables propeller and motor efficiency predictions across sweeps of U_∞ , RPM, and fixed voltages; see Figure 2 [2]. While remarkably useful, QPROP lacks the ability to determine battery current or adjust the input voltage automatically across a given mission.

The software described in this paper, UAVDEX, seeks to address the gap in open-source design tools by enabling electric aircraft performance calculations directly from selected COTS components across velocity-altitude-runtime-throttle design spaces or along specified mission profiles following the input/output map shown in Figure 3. Note that altitude (h) is used to determine ρ , and runtime (t) can be used for steady state conditions to find SOC and, therefore, V_{oc} . Manufacturers of hobbyist components are infamous for their poor documentation, meaning the UAVDEX propulsion model must aim to provide the highest possible fidelity using only the standard 3-parameter motor constants (speed constant K_V , no load current I_0 , motor resistance R_m) and constant battery resistances (R_b) [3, 4]. The UAVDEX library is implemented in Python allowing the underlying propulsion model to be linked to other common open-source

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[‡]<https://www.rcgroups.com/forums/showthread.php?606703-e-flight-calculators-compilation>

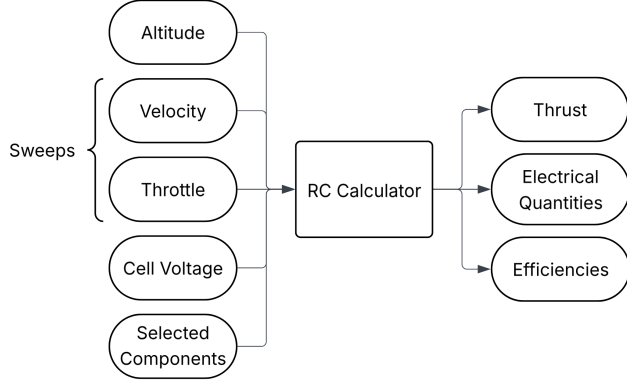


Fig. 1 RC calculator input/output diagram.

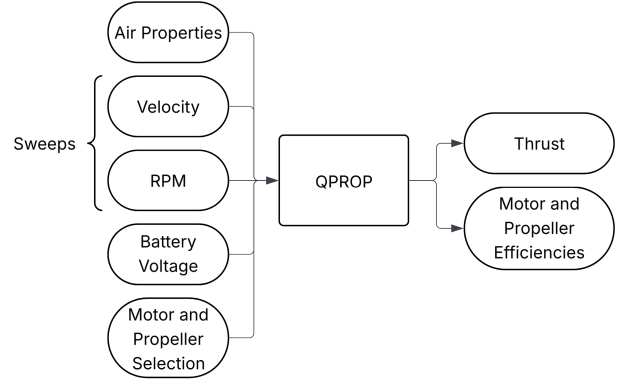


Fig. 2 QPROP input/output diagram.

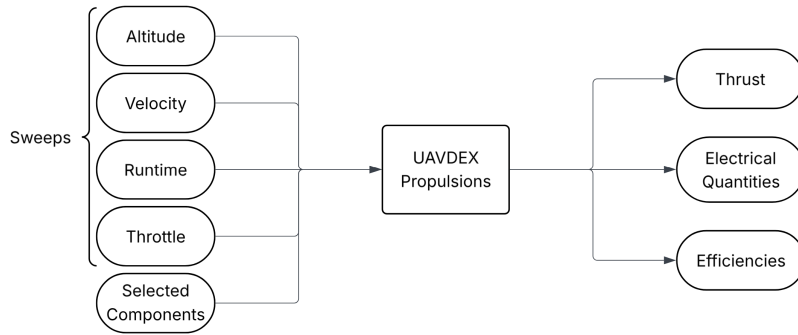


Fig. 3 UAVDEX propulsion input/output diagram.

optimization frameworks like OpenMDAO [5]. UAVDEX works best in conjunction with existing open-source tools like OpenVSP and XFLR5, which provide geometry modeling, and initial aerodynamic and stability analyses [6, 7].

The literature for electrical propulsion models and related case studies is extensive, so we focus our review on the contributions most relevant to UAVDEX and its target application. Lundström et al. describe a framework for micro air vehicle (MAV) optimization that mimics the desired workflow with specified components [8]. However, its use is restricted to experimentally characterized motors and electronic speed controllers (ESCs), and the tool is unavailable online. Gong et al. introduce an experimentally characterized model suitable for small fixed-wing electric UAVs [9]. Jeong et al. implement a multicopter-focused propulsion model into OpenMDAO alongside experimental validation [10]. Saemi and Benedict build on the work in [9] with a model that only requires the standard 3-parameter motor constants, which circumvents the frequently poor documentation by component manufacturers [3].

Saemi and Benedict’s model is intended to be used in a workflow where a conceptual sizing code, like SUAVE [11], is first used to estimate propulsion parameters. Their implementation requires known RPM, motor torque (Q_m), and battery voltage (V_b) in order to determine the full drive efficiency (η_{drive}). With propeller data, propeller torque (Q_p) can be determined from RPM, leaving two required inputs, the same as QPROP. In the original model of Gong et al., drive efficiency was determined across an RPM- Q design space. While these RPM- V_b and RPM- Q design spaces are useful, they do not fulfill the direct goal of the electric UAV designer: determining T , η_{drive} , and I_b directly from selected components and flight conditions.

UAVDEX* uses a lower-fidelity model, called SimpleRPM, that can be formulated as a single nonlinear residual with closure equations provided from performance conditions and database-specified COTS components. This paper begins with an overview of the UAVDEX library alongside a detailed explanation of the electric propulsion modeling. Subsequently, validation of the SimpleRPM model against a variety of experimental data is presented. Finally, a case study demonstrating application of UAVDEX for the design of RPI DBF’s 6th place aircraft is detailed.

*<https://github.com/sammy183/UAVDEX>

II. Methodology

A. Preliminaries

Range for an electric aircraft in level flight with a set battery is estimated by

$$R \text{ (in m)} = 3600 \frac{L}{D} \eta_{\text{drive}} \frac{k E_{\text{batt}}}{W}$$

where L/D is the aircraft's aerodynamic efficiency, η_{drive} is the efficiency of the entire propulsion drive, E_{batt} is the battery energy in Wh, and W is the aircraft weight in newtons. The coefficient k is a knockdown factor based on depth of discharge (DOD), cell efficiency, and capacity fade near max DOD [12]. The RPI DBF team typically sets $k = 0.75$ based on previous flight experience. This is an optimistic estimate but has proven useful for competition designs necessitating tight margins.

Endurance for an electric aircraft is the minimum of the level flight endurance or the power system specific endurance, which can be estimated with

$$t_{\text{level}} \text{ (in s)} = 3600 \frac{L}{D} \eta_{\text{drive}} \frac{k E_{\text{batt}}}{W U_{\infty}}$$

$$\text{and } t_{\text{sys}} \text{ (in s)} = \frac{3.6 C_B}{I_b}$$

respectively, where U_{∞} is the level flight velocity in m/s, C_B is the battery capacity in mAh, and I_b is the battery current in amps, acting similarly to the fuel flow rate in a conventional aircraft. Note that t_{sys} is often the limiting endurance term.

To be useful to the designer, an effective tool for multidisciplinary design analysis (MDA) of electric aircraft must provide rapid calculation of T , I_b , and η_{drive} across varying flight conditions defined by U_{∞} , δT , ρ , and V_{oc} . Additionally, quantities like motor power ($P_{in,m}$) and ESC current (I_c) are often needed to provide constraints on the system.

B. UAVDEX Overview

At a high level, UAVDEX provides propulsion, mission, and optimization modules to the designer. Figure 4 shows UAVDEX's functionalities for MDA with related inputs and underlying models. This work will focus on the propulsion analysis functionality.

After defining motor, battery, and propeller components from COTS databases, propulsion analyses can be used to determine drive characteristics across the velocity-altitude-runtime-throttle design space. These characteristics include efficiencies, useful to determine the optimal flight conditions, and electrical metrics such as battery current and motor electric power, which provide important constraints.

UAVDEX is designed to be used without necessitating more than basic Python knowledge. The COTS databases, given as CSVs, are easily adjusted using spreadsheet software (e.g. Excel). This is a key attribute for accessibility: many RC calculators are useless to the modern designer as their databases are outdated, or their website-based links have broken. With freely available spreadsheet databases, users with limited technical knowledge can add their own propulsion components. Additionally, any common units can be used with UAVDEX, meaning usage is intuitive for students around the world. Finally, UAVDEX leverages the Numba open-source just-in-time compiler to accelerate its runtime for multidimensional design space investigations [13].

Due to the poor and inaccurate documentation of hobbyist electrical components, UAVDEX assumes constant values for ESC efficiency (η_c) and no-load current (I_0) by default. If an experimental characterization of I_0 exists in the database — generated via the methodology presented in [4] — UAVDEX can use that to improve prediction accuracy (see Section III.B.3 for an example). By default, propeller aerodynamic coefficients are determined from precomputed blade element momentum theory (BEMT) codes provided by APC propellers [14]; however, experimental data, primarily sourced from the UIUC propeller database, is used when available [15–19]. The APC BEMT data is required because of a gap in available propeller experimental data, specifically for higher-performance setups producing >1000 W of mechanical power.

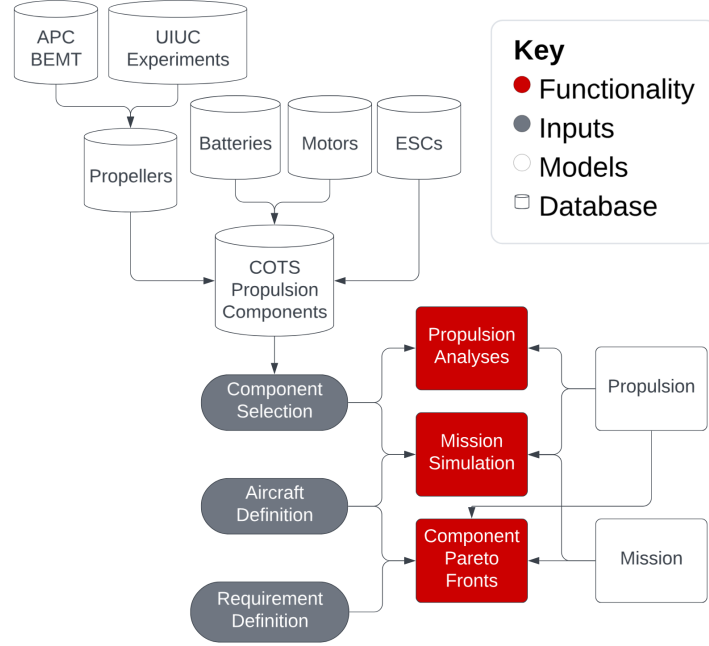


Fig. 4 UAVDEX MDA functionality structure.

C. Propulsion Modeling

The power and efficiency chain for a modern electric UAV is shown in Figure 5. A lithium-polymer (LiPo) or lithium-ion (Li-ion) battery feeds DC current (I_c) into one or more ESCs that output three-phase AC current (I_m) into brushless DC (BLDC) motors. The motors spin propellers through direct drive or gear reduction.

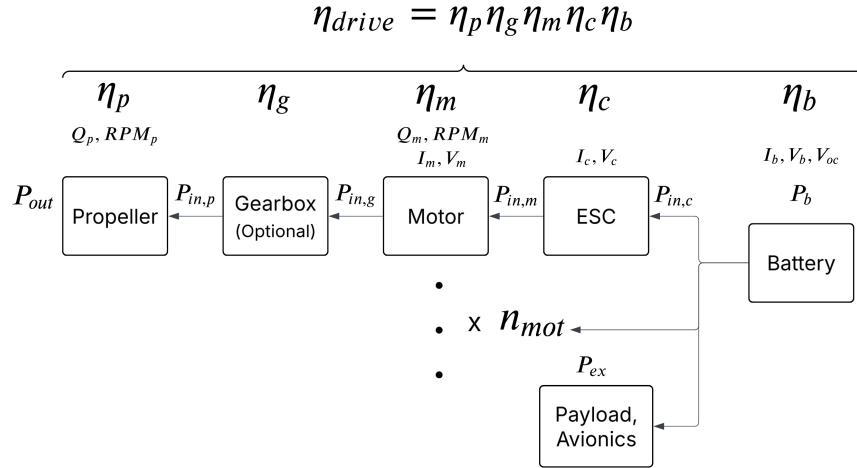


Fig. 5 Power and efficiency chain.

The SimpleRPM model represents this system as a single nonlinear equation by assuming a constant η_c and I_0 , and by obtaining propeller coefficients as functions of RPM_p and advance ratio, J , from BEMT or experimental data. For pedagogical clarity, every sub-equation will be detailed in this section. Readers familiar with the underlying theory and equations can proceed to the final nonlinear residual provided in Equation (5).

The model starts with the propeller and assumes an input RPM_p . Initially, it inputs the average of the lower and upper RPM bounds from the propeller data. The torque created by the propeller is determined as a function of the

selected setup and design space/mission profile variables through the following sequence of equations:

$$\begin{aligned} n &= \frac{\text{RPM}_p}{60}, \\ J &= \frac{U_\infty}{nD}, \\ C_P &= C_P(\text{RPM}_p, J), \\ C_Q &= \frac{C_P}{2\pi}, \\ Q_p &= \rho(h)n^2D^5C_Q, \end{aligned}$$

where n is revolutions per second, C_Q is the propeller torque coefficient, Q_p is the propeller torque, D is the propeller diameter, and C_P is the propeller power coefficient. D and the power coefficient function $C_P(\text{RPM}_p, J)$ are obtained from the propeller component data. The design variables are U_∞ , freestream velocity, and h , altitude. By default, the density, $\rho(h)$, is determined from altitude using the 1976 U.S. Standard Atmosphere [20], but alternatives, such as the climate specific atmospheres from MIL-STD-210 [21] or a manually input air density, are also available.

Gear reduction allows high K_V inrunner motors to spin larger diameter propellers. This can be advantageous since larger diameter fixed-pitch propellers tend to be more efficient while inrunner motors have a smaller form factor, which reduces frontal area. At small scale, planetary gearboxes comprise the vast majority of commercial offerings, with efficiencies between 92-96% [22]. The conversions between propeller and motor RPMs and torques are given by

$$\begin{aligned} Q_m &= \frac{Q_p}{\text{GR} \eta_g}, \\ \text{RPM}_m &= \text{RPM}_p \text{ GR}, \end{aligned}$$

where GR is the gear ratio from component specifications, and η_g is the gearbox efficiency, set to 94% [22] by default.

A BLDC motor at steady-state is approximated by the diagram shown in Figure 6. For most flight segments, steady-state analyses are appropriate for UAV propulsion systems since the transient pulses last milliseconds, so their duration is negligible in the context of a full flight [23].

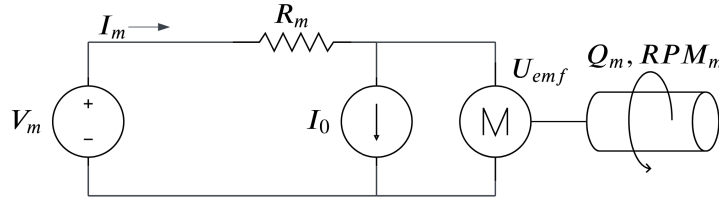


Fig. 6 Brushless DC motor diagram.

The motor current (I_m) is determined by

$$I_m = Q_m K_V \frac{\pi}{30} + I_0,$$

where K_V is the motor speed constant in RPM/V, and the $\frac{\pi}{30}$ factor converts K_V to rad/s/V. K_V in RPM/V is preferred to the SI equivalent because it enables direct use of hobbyist manufacturer specification sheets, thereby increasing accessibility. As established in the literature, I_0 varies linearly with RPM and the motor voltage V_m [4, 8, 9]. A fit of I_0 versus V_m is easier to measure experimentally, requiring only a power monitor, but a fit of I_0 versus RPM can be used directly in the single nonlinear residual without any additional equations. Approximating I_0 from the manufacturer value tends to result in increased error for propulsion systems with lower power draw. However, validation in Section III.B.3 indicates that the I_0 error is relatively minimal.

Once I_m has been estimated, ESC current (I_c) is obtained via

$$I_c = \frac{\delta T I_m}{\eta_c}$$

by assuming a constant, default ESC efficiency of $\eta_c = 93\%$ based on experimental results [3, 8, 10]. In general, η_c varies with V_b , I_c , and throttle setting (δT). Gong et al. [9] found η_c between 86-96% while Saemi and Benedict [3] found η_c between 90-100%. Lundström et al [8], working with older ESCs, found that for δT greater than 40%, η_c ranged nonlinearly between ~ 65 -100%. One consistent result is that η_c was maximized at 100% δT , indicating that electric UAVs should be designed to cruise close to full throttle for higher η_{drive} and thereby optimal range.

Battery current (I_b) is obtained by simply multiplying ESC current by the number of motors (n_{mot}):

$$I_b = I_c n_{\text{mot}}.$$

For analysis of component-specified designs at steady-state conditions over some runtime, t , the battery SOC is determined by

$$\text{SOC} = 1.0 - \frac{\int_0^t I_b(\tau) d\tau}{3.6 C_B n_p}, \quad (1)$$

where C_B is the battery capacity in mAh and n_p is the number of cells wired in parallel. For propulsion specific analyses, assuming constant current reduces the integral in Equation (1) to $I_b \cdot t$, allowing for propulsion quantity plots with respect to runtime. When analyzing an aircraft over a discrete mission profile, I_b is found for different flight conditions, and then the time spent in each condition is used to find the final SOC.

Open-circuit voltage (V_{oc}) is a function of SOC and battery health. For LiPo batteries, which account for the majority of modern small UAV powerpacks, the cell discharge curve can be estimated using [10]

$$V_{oc}(\text{SOC}) = 1.7 \text{ SOC}^3 - 2.1 \text{ SOC}^2 + 1.2 \text{ SOC} + 3.4.$$

In Section III.B.2, the above battery model from [10] is compared to that from [24] and experimental data.

With known V_{oc} and I_b , battery voltage (V_b) is calculated by

$$V_b = n_s V_{oc} - I_b R_b, \quad (2)$$

where n_s is the number of cells wired in series.

Working back up the power chain from the battery, voltage in the motor is controlled by the ESC, which uses Pulse-Width Modulation (PWM) to transfer an average voltage to the motor. This average matches δT times the DC battery voltage, V_b [25]. From the motor voltage (V_m) and current (I_m), RPM_m is determined via Equation (3), listed below. Note that K_V is again expressed in RPM/V. Finally, RPM_m is converted via the gearbox to RPM_p in Equation (4), also listed below.

$$V_m = \delta T V_b$$

$$\text{RPM}_m = K_V (V_m - I_m R_m) \quad (3)$$

$$\text{RPM}_p = \frac{\text{RPM}_m}{\text{GR}} \quad (4)$$

To summarize, the nonlinear equation for RPM_p can be expressed succinctly as

$$\text{Res}(\text{RPM}_p) = \text{RPM}_p - \widehat{\text{RPM}_p}(\text{RPM}_p) \quad (5)$$

where the function $\widehat{\text{RPM}_p}$ is given by

$$\widehat{\text{RPM}_p} = \frac{K_V (\delta T V_b - I_m R_m)}{\text{GR}} \quad (6)$$

with

$$n = \frac{\text{RPM}_p}{60} \quad J = \frac{U_\infty}{nD}$$

$$Q_m = \frac{\rho(h) n^2 D^5 C_P (\text{RPM}_p, J)}{2\pi \text{GR} \eta_g} \quad I_m = Q_m K_V \frac{\pi}{30} + I_0$$

$$I_b = \frac{\delta T I_m n_{\text{mot}}}{\eta_c} \quad \text{SOC} = 1.0 - \frac{\int_0^t I_b(t) dt}{3.6 C_B n_p}$$

$$V_b = n_s V_{oc}(\text{SOC}) - I_b R_b$$

In the above equations, the constants D , GR , K_V , I_0 , C_B , n_p , n_s , R_b , and R_m are found via the database entries for selected components, all of which are commonly available online. U_∞ , h , t , and δT are the design variables whose values or ranges are defined by the user.

Frequently, SOC or V_{oc} are used directly as the design space variable instead of t . When flying an RC aircraft, the pilot often uses V_{oc} as the primary metric to determine when to land. Nevertheless, explicit knowledge of how characteristics vary over t is often more useful to the conceptual designer. Current varies significantly in flight maneuvers, meaning that for aircraft primarily flying a steady cruise condition, the constant I_b assumption is acceptably accurate.

Equation (5) is easily solved by an implementation of the secant method with bisection as a safeguard. Once Equation (5) is solved for RPM_p , efficiencies and thrust are determined via,

$$\eta_m = \frac{P_{in,c}}{P_{in,m}} \quad \eta_p = \frac{C_T J}{C_P} \quad \eta_b = 1.0 - \frac{I_b^2 R_b}{n_{\text{mot}} P_{in,c} + I_b^2 R_b}$$

$$\eta_{\text{drive}} = \eta_c \eta_m \eta_g \eta_p \eta_b \quad T = n_{\text{mot}} \rho n^2 D^4 C_T(\text{RPM}_p, J)$$

where

$$P_{out} = \frac{\pi}{30} \text{RPM}_p Q_p \quad P_{in,m} = V_m I_m \quad P_{in,c} = V_b I_c$$

A limitation of this model is neglecting the variation of battery resistance with SOC and temperature. Determining how hobbyist components transfer heat is often impractical due to insufficient documentation and a variety of installation techniques in UAV airframes.

III. Validation

A. Setup

The SimpleRPM model was validated using static thrust testing and in-flight telemetry from a small electric UAV. This section describes the experimental setup for these validations.

In static conditions, 7 APC propellers with experimental data available from UIUC [26] were tested using Cobra C-4130/20 and SunnySky V8110-170 BLDC motors and Gaoneng 8S 3300mAh LiPo batteries. Thrust, torque, battery current, and battery voltage were measured on a custom thrust stand shown in Figure 7. RPM and throttle setting were determined by telemetry from a Spektrum Avian 80A ESC. A temperature and humidity sensor was used before and after each test to determine the air density.

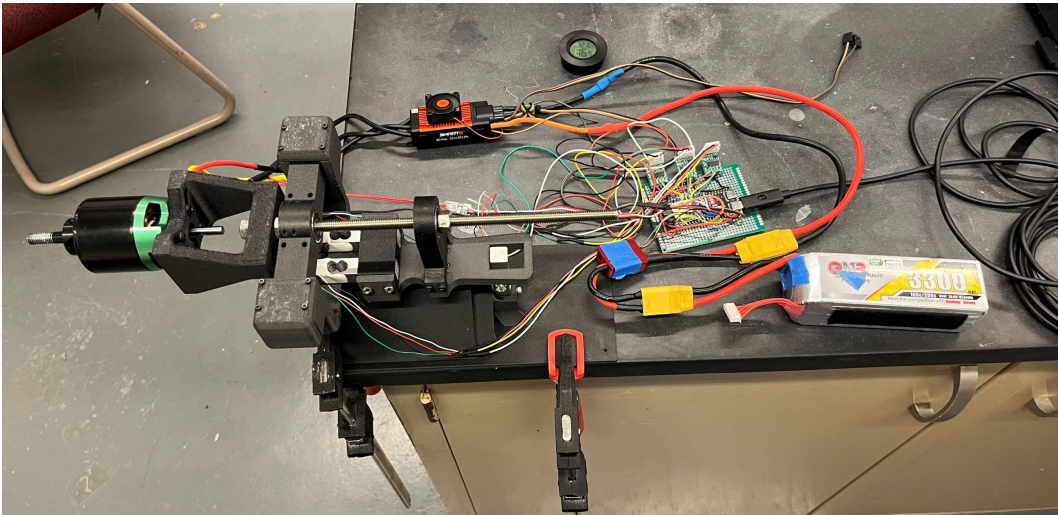


Fig. 7 Cobra C-4130/20 motor on thrust test stand connected to a Spektrum Avian 80A ESC, ESP32 controller, and Gaoneng 8S1P 3300 mAh LiPo battery.

Using a Matek F405 Wing flight controller, pitot tube, power monitor, and PWM passthrough, a prototype aircraft from the 2026 RPI DBF team (Figure 8) gathered flight data consisting of air pressure, temperature, freestream velocity, throttle input, RPM, battery current, and battery voltage. With known ρ , U_∞ , δT , and V_{oc} , the SimpleRPM outputs were compared directly to experimental RPM, I_b , and V_b .



Fig. 8 Flight test prototype used for propulsion validation.

B. Static Test Results

1. Propeller Coefficients

To confirm the validity of the experimental setup, C_P and C_T values for APC 18x10E, 14x8.5E, 16x10E, 16x12E, 18x12E, 20x10E, and 18x8E propellers were compared directly to APC BEMT and UIUC data in Figures 9 and 10. The static experimental data was gathered using a stepping procedure containing 7 intervals from 30-100% δT . The steps were held for 10 seconds each. After filtering outliers from the interquartile range (IQR) for a given throttle level, the average was taken. The filtering procedure was adopted to remove transient characteristics at the beginning and end of each step.

Table 1 shows that the RPI experiments matched the UIUC data within 10%. Due to time constraints, each run was replicated only two or three times, resulting in a larger than ideal variance. Some of the propellers, like the 18x8E, displayed a significant overprediction in C_T and C_P compared to the UIUC data. The propellers used in this experiment were representative of frequent real-world usage, having flown on a number of past RPI DBF aircraft. This may have resulted in a significant geometry difference between the two propellers. Additionally, the plots in Figures 15 and 16 demonstrate that the UIUC data does not extend to higher power setups (i.e. the 18x10E results), necessitating use of the APC BEMT data for design competitions like DBF.

However, from Figures 9 and 10, the APC BEMT data significantly underpredicts C_T and C_P for static conditions, especially for larger propellers like the 18x8E, 18x10E, and 20x10E. This was examined further by comparing APC BEMT for 76 propellers to their UIUC experimental data across static and dynamic conditions. The relative C_T and C_P errors are shown in Figures 11a and 11b. The data in these figures indicates a median error of 15% in thrust from the APC C_T data. The C_P median error of 10% will correspondingly induce an error in the battery current prediction.

Table 1 Combined relative errors for C_T and C_P .

Comparison	Mean C_T RE (%)	Median C_T RE (%)	Mean C_P RE (%)	Median C_P RE (%)
RPI vs UIUC	9.76	8.90	9.08	6.49
UIUC vs BEMT	14.31	13.99	10.00	9.13
RPI vs BEMT	22.25	29.30	36.08	21.55

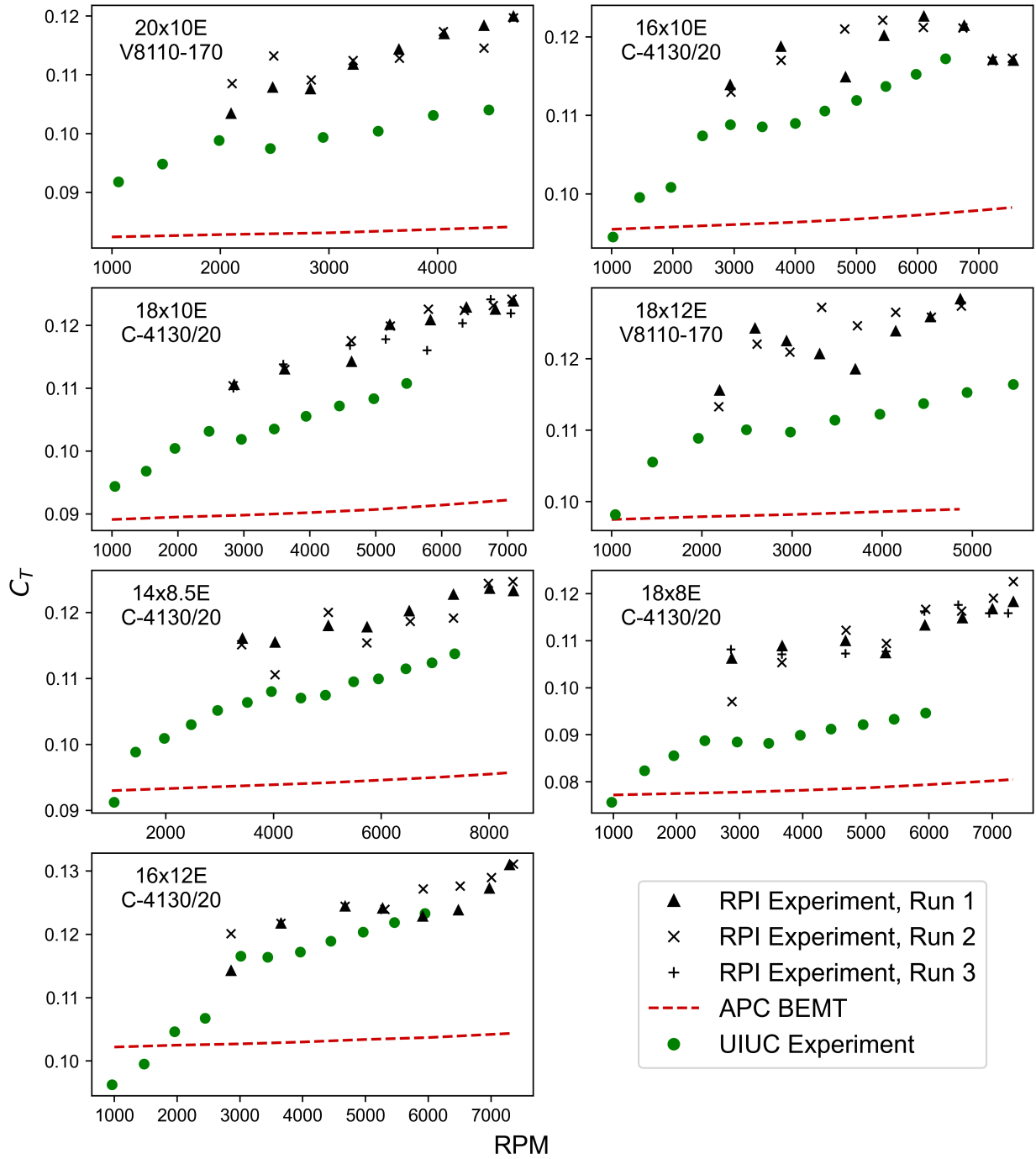


Fig. 9 C_T for experiments vs APC BEMT data and UIUC experimental data.

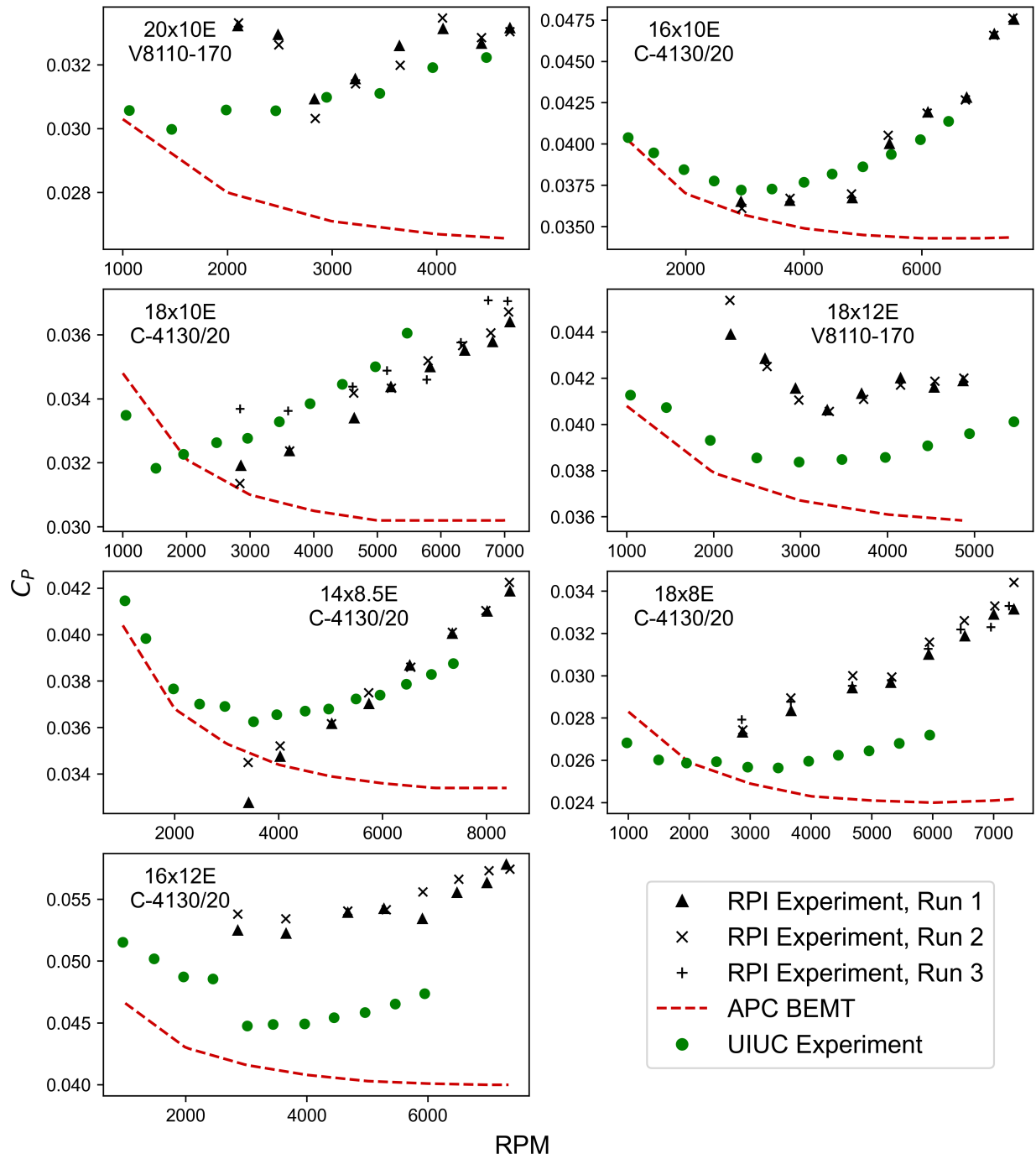


Fig. 10 C_P for experiments vs APC BEMT data and UIUC experimental data.

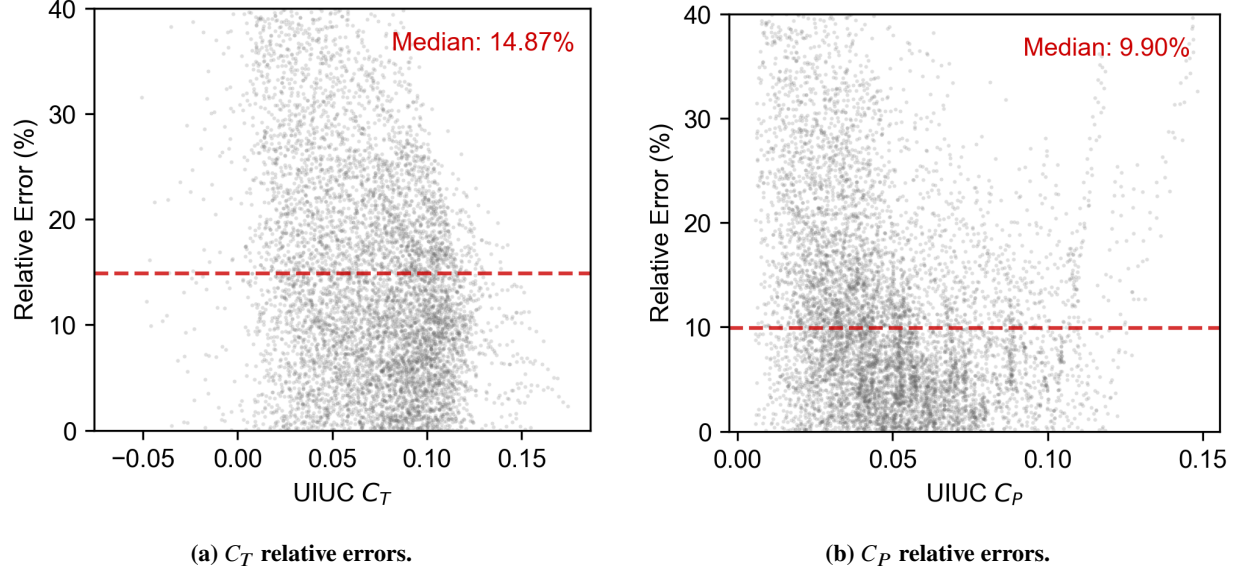


Fig. 11 Relative errors of APC BEMT vs UIUC data for 76 propellers.

2. Battery Modeling

Figure 12 compares a full discharge test of the Gaoneng 8S1P 3300 mAh battery against LiPo battery models from Jeong et al. [10] and Chen et al. [24]. V_{oc} was calculated from experimental V_b , I_b and Equation (2) using R_b s taken experimentally from a battery charger indicator. Chen et al.'s model had an average relative error of 2.19% and a maximum relative error of 9.27%. Jeong et al.'s model had an average relative error of 1.19% and a maximum relative error of 4.06% although it lacked the characteristic sharp peak near full charge. Jeong et al.'s battery model is set by default in UAVDEX.

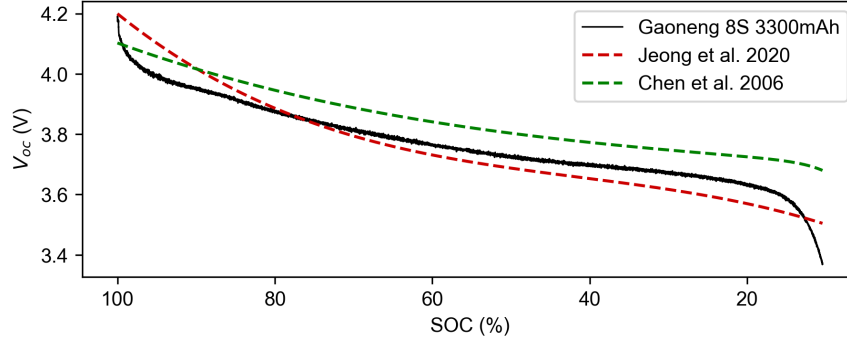


Fig. 12 LiPo battery model comparison to experiment.

3. Constant I_0

A linear fit of I_0 vs RPM was determined experimentally for both the Cobra and Sunnysky motors (Figures 13 and 14). These linear fits were used alongside the experimental results for Q_m , RPM_m , and δT to determine I_b . Additionally, I_b was calculated using the original equipment manufacturer (OEM) I_0 values. The absolute error between I_b values calculated using the linear fit of I_0 and constant values of I_0 is shown in Figure 15. The vast majority of the error depended only upon the motor used and accounted for less than 1 A in all tested cases. At worst this resulted in a 5% difference in I_b for the SunnySky 18x12E test at low throttles, which is insignificant compared to the error from the APC BEMT C_P values. Therefore, the constant I_0 assumption is a worthwhile tradeoff to allow prediction of hobbyist motors lacking experimental characterization, which is the majority of motors in this context!

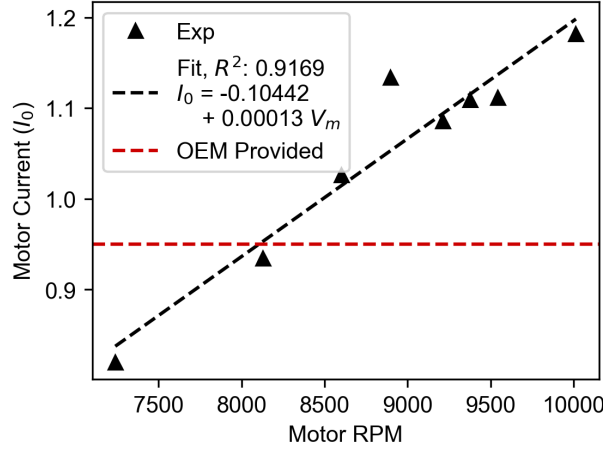


Fig. 13 I_0 fit for Cobra C-4130/20 motor.

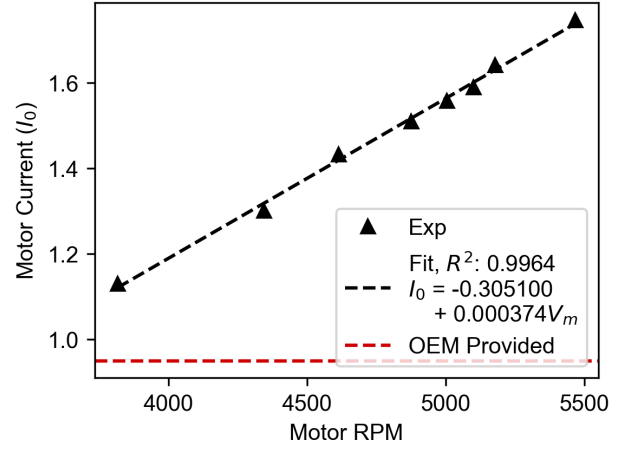


Fig. 14 I_0 fit for SunnySky V8110-170 motor.

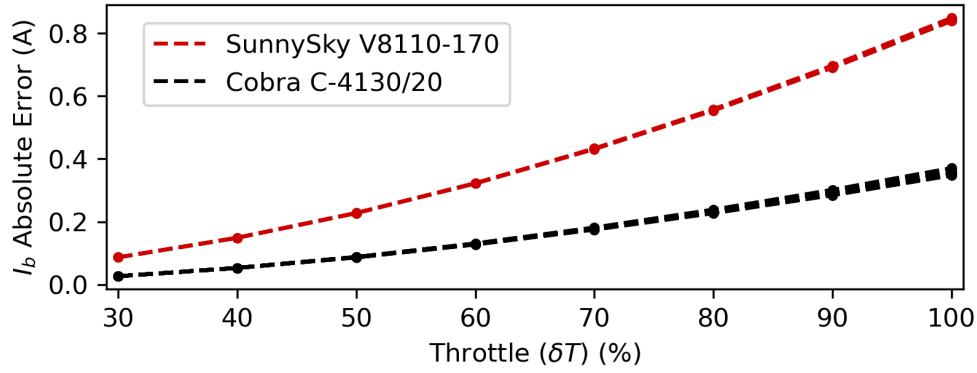


Fig. 15 Error for I_b calculated with constant values vs linear fits of I_0 for all runs.

4. Full Model Comparison

Figures 16, 17, and 18 compare static test results against predictions from the SimpleRPM model and a popular RC calculator named eCalc [27]. Both the SimpleRPM and eCalc inputs for ρ , δT , V_{oc} , and $U_\infty = 0$ were taken directly from the experimental data for each run.

The relative and absolute errors are given in Table 2. The SimpleRPM model produced acceptable error ($\leq 15\%$) for I_b and particularly low error ($\leq 5\%$) for RPM. However, both the SimpleRPM model and eCalc displayed unacceptable underprediction of thrust with around 25% error.

While both the SimpleRPM model and eCalc had similar levels of error for T and I_b , eCalc tended to significantly underpredict RPM (Figure 17).

To investigate the impact of the significant BEMT C_T error, SimpleRPM thrust results were recalculated using experimental C_T values and the results are plotted in Figure 19. The substitution of experimental C_T values produced a more reasonable error of around 10%, indicating that the underlying model is acceptable for early stage design if sufficiently accurate C_T data is available. Unfortunately, due to the aforementioned gap in propeller data, this limits the usefulness of this tool for higher power setups, capping it at around 25% median accuracy for static conditions. At lower power setups using C_T data from the UIUC database, the accuracy was around 15%.

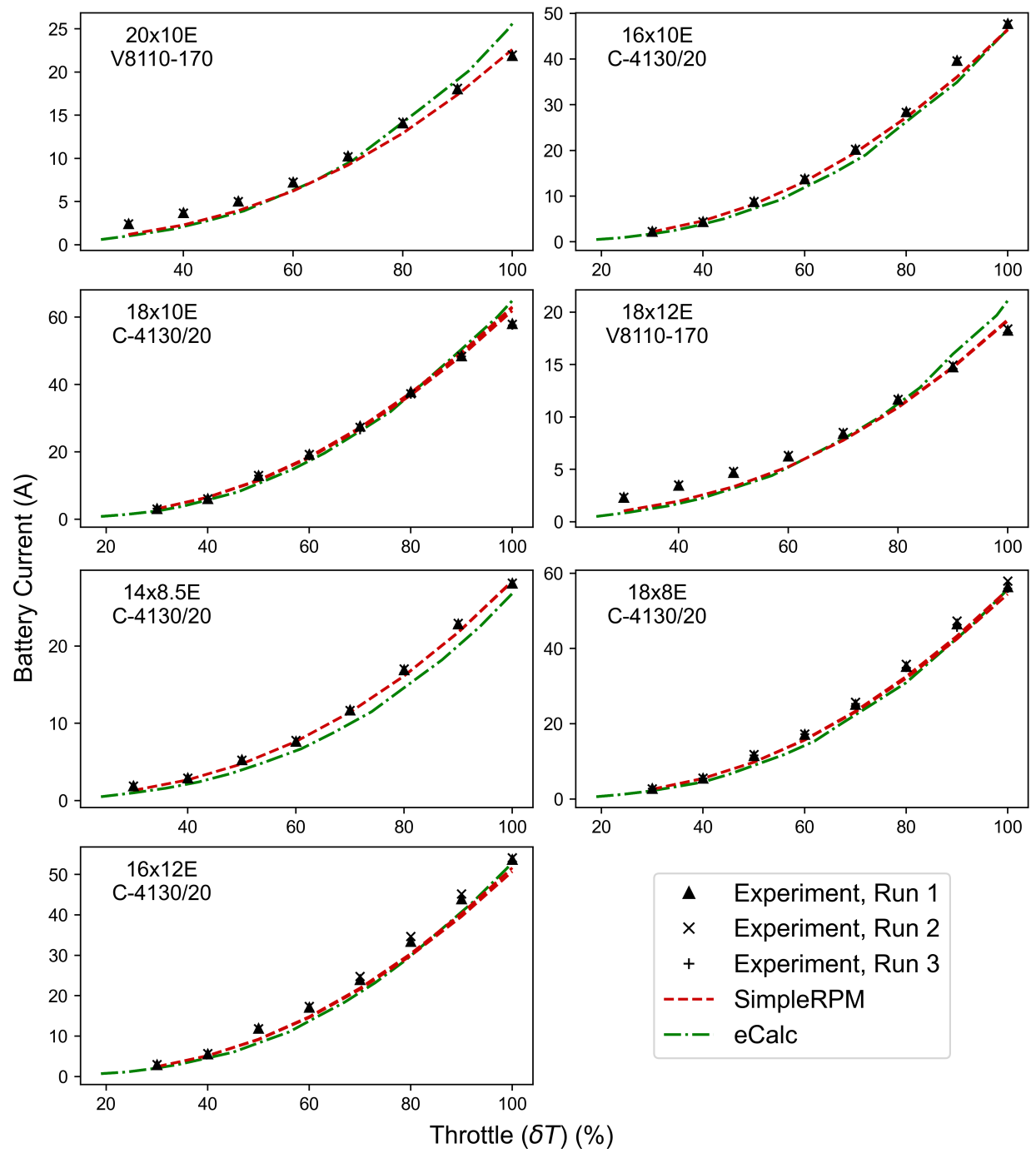


Fig. 16 I_b vs δT for SimpleRPM model against experiment.

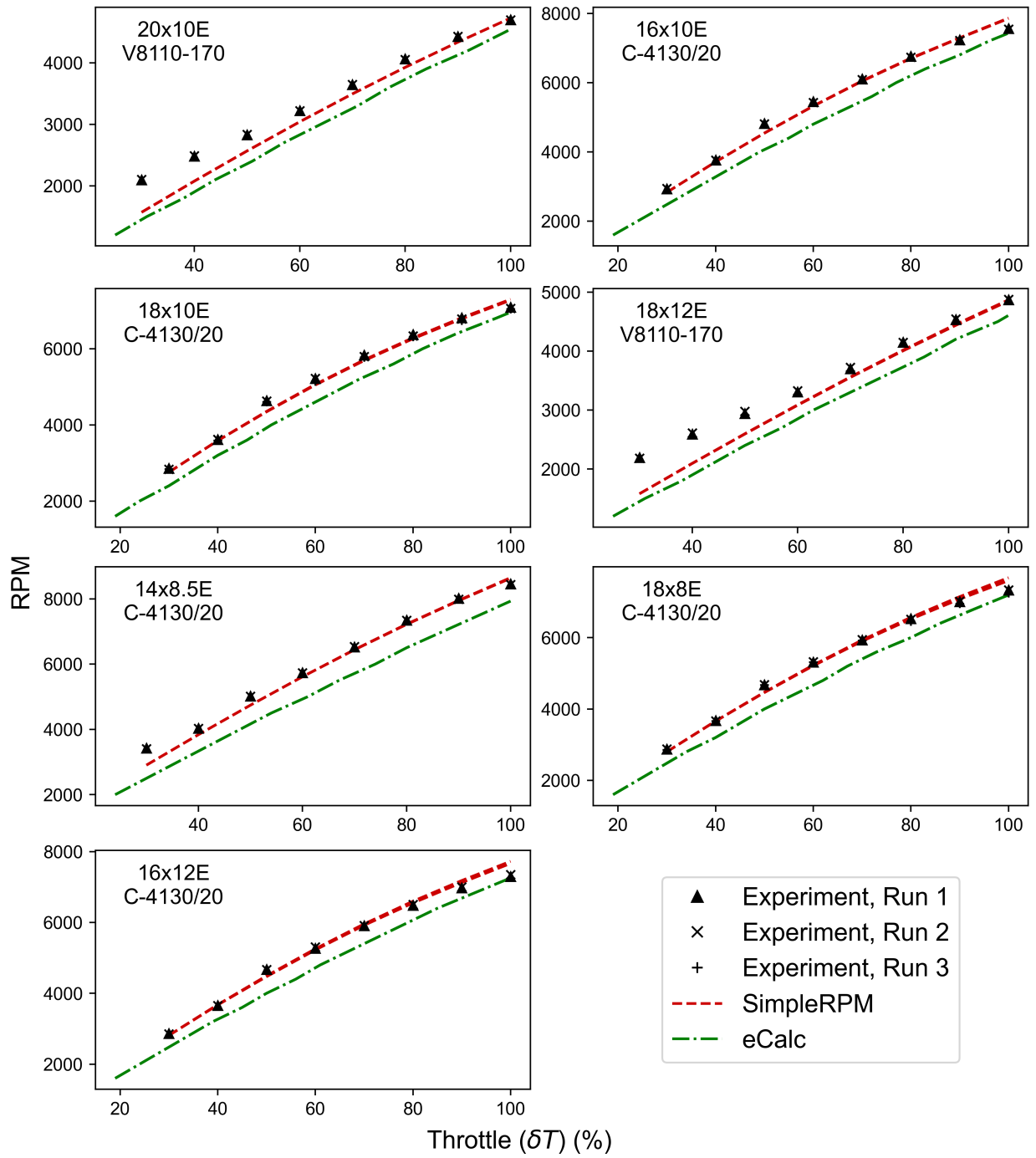


Fig. 17 RPM vs δT for SimpleRPM model against experiment.

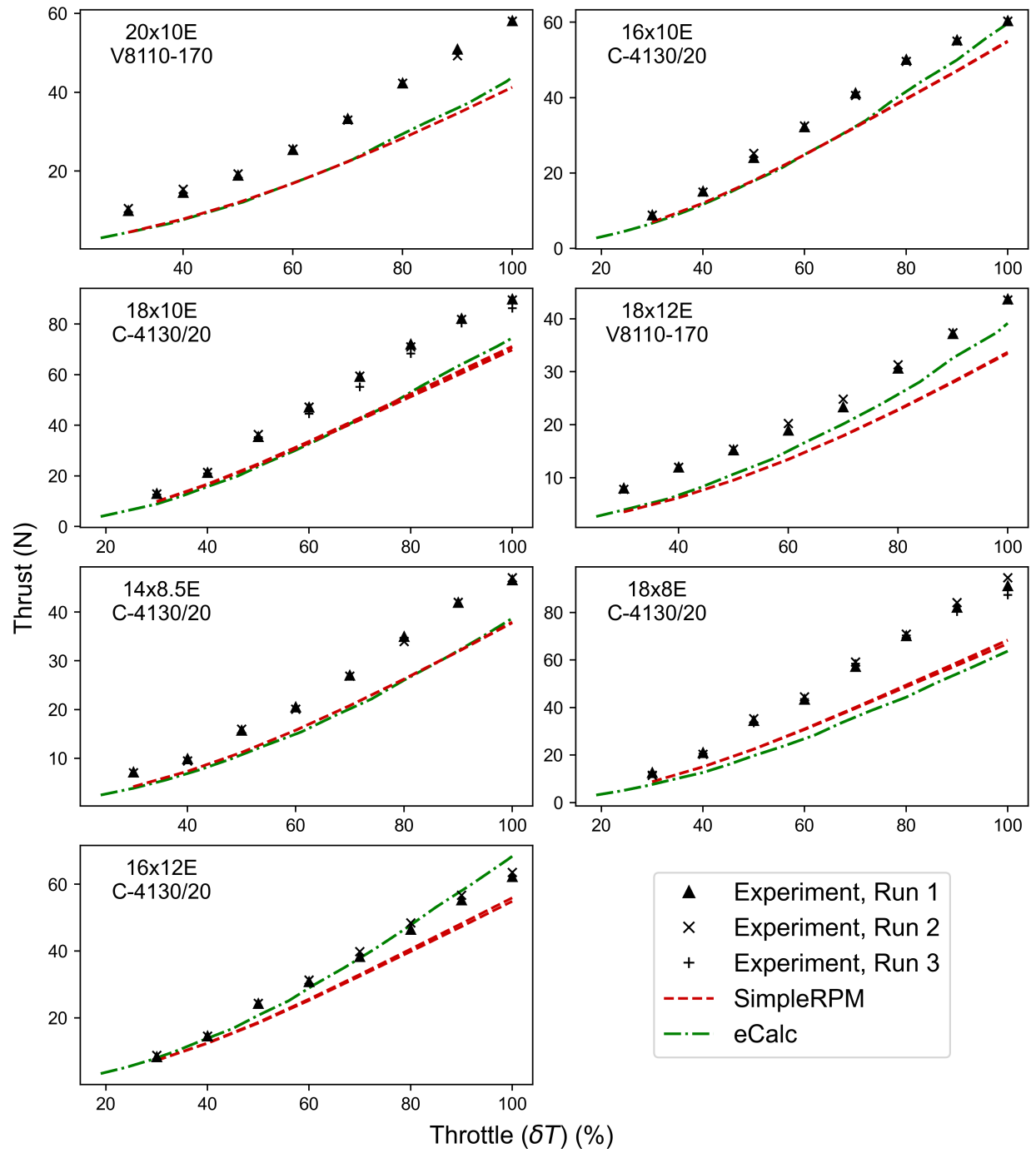


Fig. 18 T vs δT for SimpleRPM model against experiment.

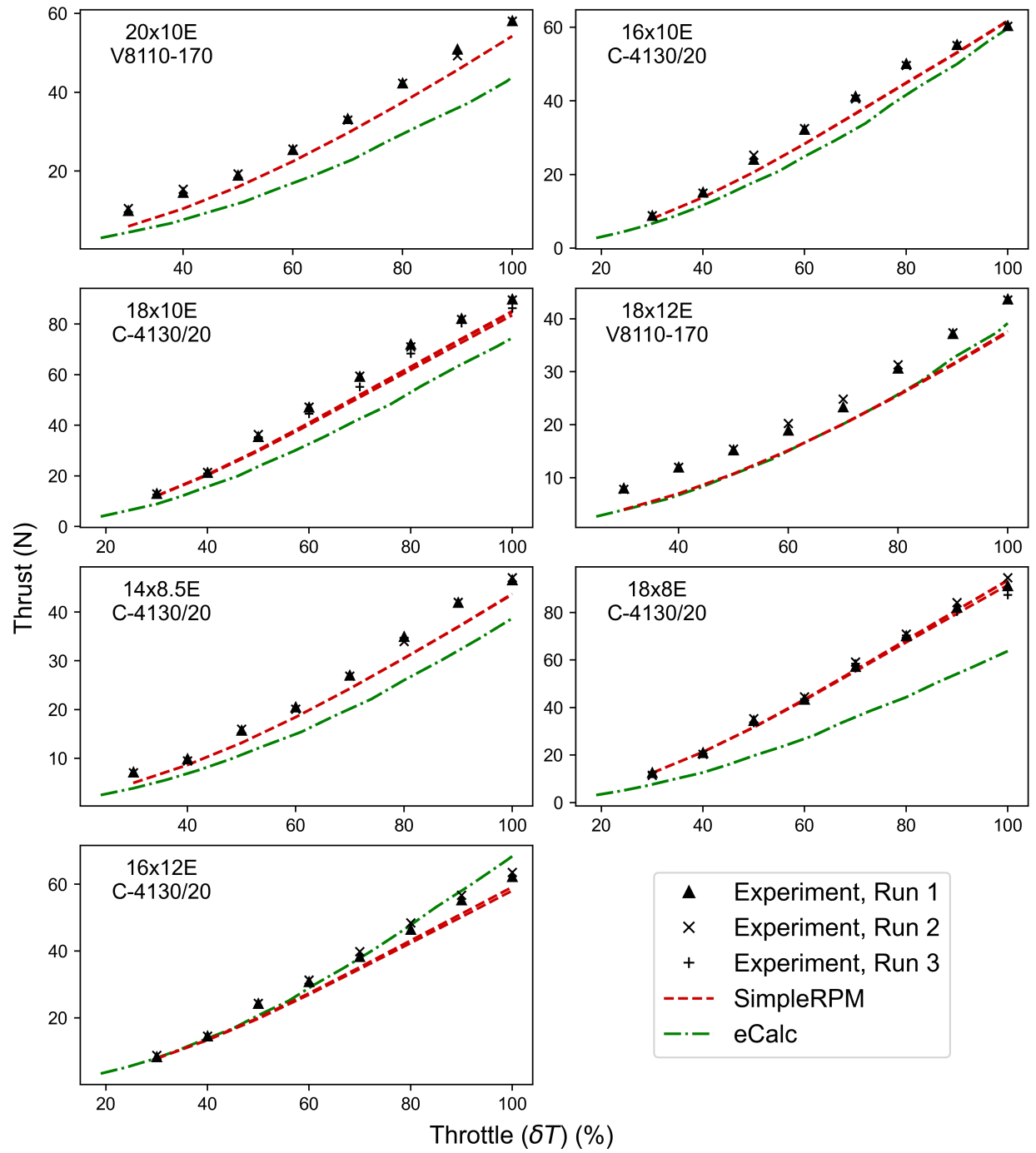


Fig. 19 T vs δT for SimpleRPM model with corrected C_T against experiment.

Table 2 Absolute error (AE) and relative error (RE) for SimpleRPM and eCalc against static tests.

Model	Mean AE	Median AE	Mean RE (%)	Median RE (%)
Battery current (I_b) (A)				
SimpleRPM	2.2910	1.7511	15.8886	13.5848
eCalc	1.9545	1.6170	16.2226	13.1975
RPM				
SimpleRPM	160.2175	114.2703	4.1375	2.2598
eCalc	498.7631	508.8768	11.4494	10.9137
Thrust (T) (N)				
SimpleRPM	9.6756	7.8128	26.8842	25.2504
eCalc	9.4363	7.2009	26.2185	26.6585
SimpleRPM (corrected C_T)	3.4630	3.6477	11.8762	9.8321

C. Flight Test Results

Figures 20, 21, and 22 compare SimpleRPM model predictions of RPM, I_b , V_b , and SOC against experimental results from three flights of an RPI DBF prototype aircraft using a Gaoneng 8S 3300 mAh battery, a SunnySky V8110-170 motor, and an APC 22x12E propeller. The data shown used Savitzky-Golay filtering [28] to alleviate sensor noise.

Relative error metrics listed in Table 3 indicate very good agreement for V_b and RPM, with good agreement for I_b in flights 2 and 3. Flight 1 displayed significant error in I_b , which may be attributed to the more extreme dynamics of that flight. For instance, in Figure 20, RPM varies between ~ 3000 – 4500 , while in the other flights RPM varies between ~ 4100 – 4400 . Despite these challenges, the ending SOC prediction only differed by 5.9% as shown in Table 4 and Figure 20. On the whole, SOC prediction using the SimpleRPM model was within the 15% accuracy target, with Flights 2 and 3 possessing SOC values that differed by only 2.3% and 1.6% at worst, respectively. These error levels are satisfactory for early stage design. The accurate in-flight RPM prediction also reinforces the necessity for high performance experimental propeller data to ensure an equally accurate thrust prediction.

Table 3 Relative error metrics (Max RE) for flight testing.

Test	Comparison	Max RE (%)	Mean RE (%)	Median RE (%)
Flight 1	RPM	22.9984	2.9816	1.9332
	I_b (A)	92.3463	24.8874	23.1419
	V_b (V)	3.1029	1.7671	1.7000
Flight 2	RPM	3.1603	0.9710	0.9039
	I_b (A)	11.1621	4.8048	4.8097
	V_b (V)	3.0193	2.7416	2.7653
Flight 3	RPM	2.9077	1.0521	0.9434
	I_b (A)	12.9388	5.1470	5.1279
	V_b (V)	3.0103	2.6449	2.6919

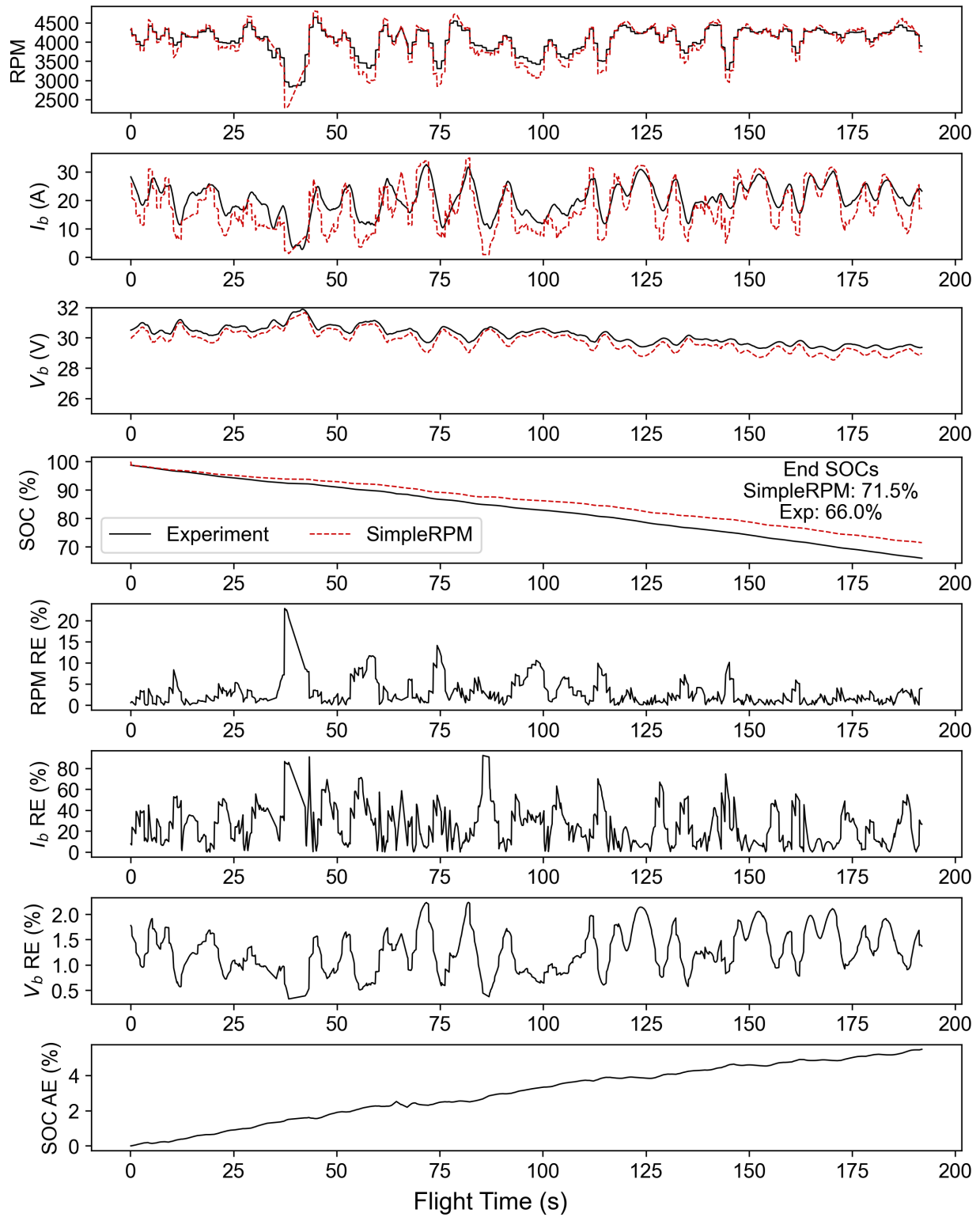


Fig. 20 Flight 1 data vs SimpleRPM with relative errors (RE) and absolute errors (AE).

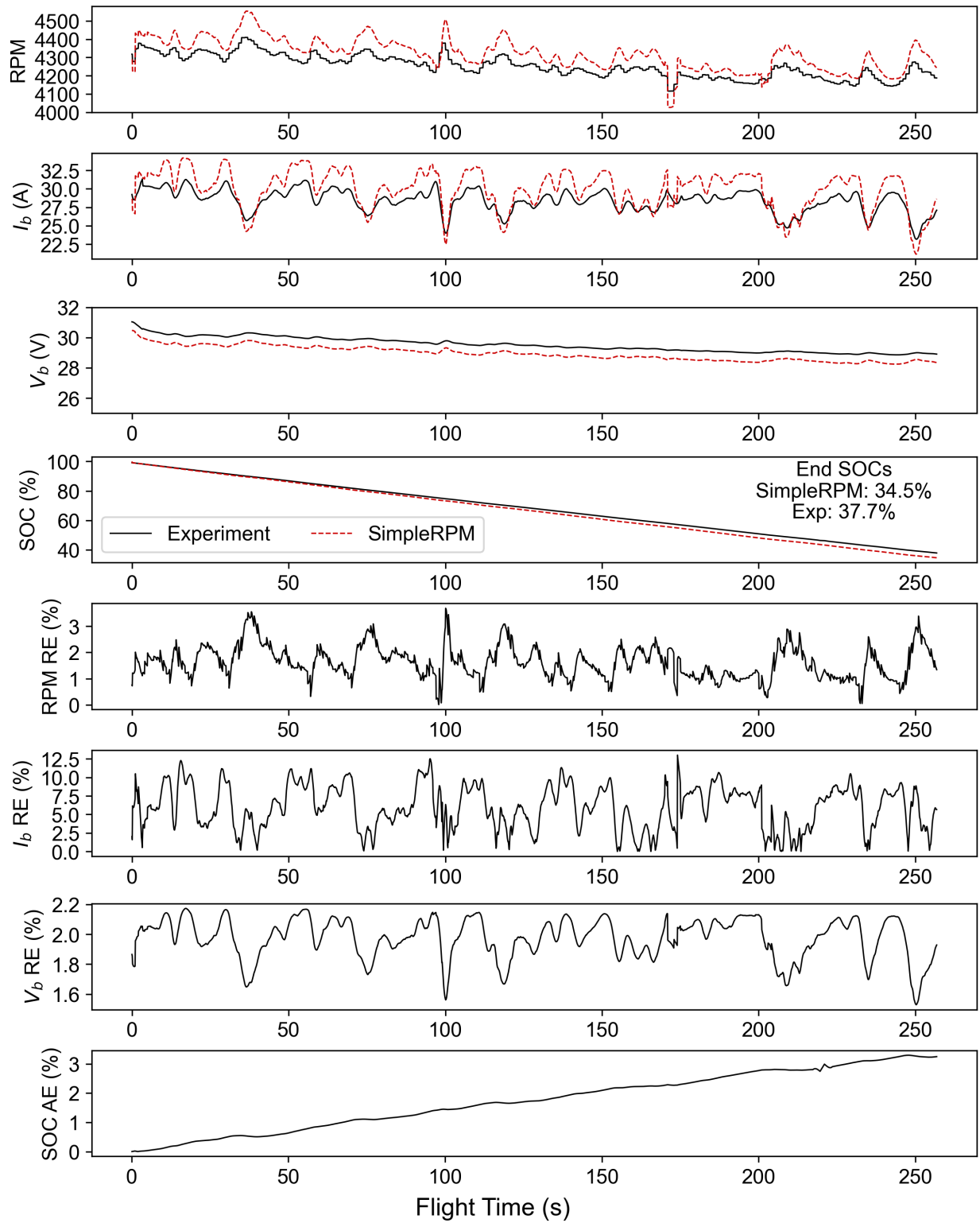


Fig. 21 Flight 2 data vs SimpleRPM with relative errors (RE) and absolute errors (AE).

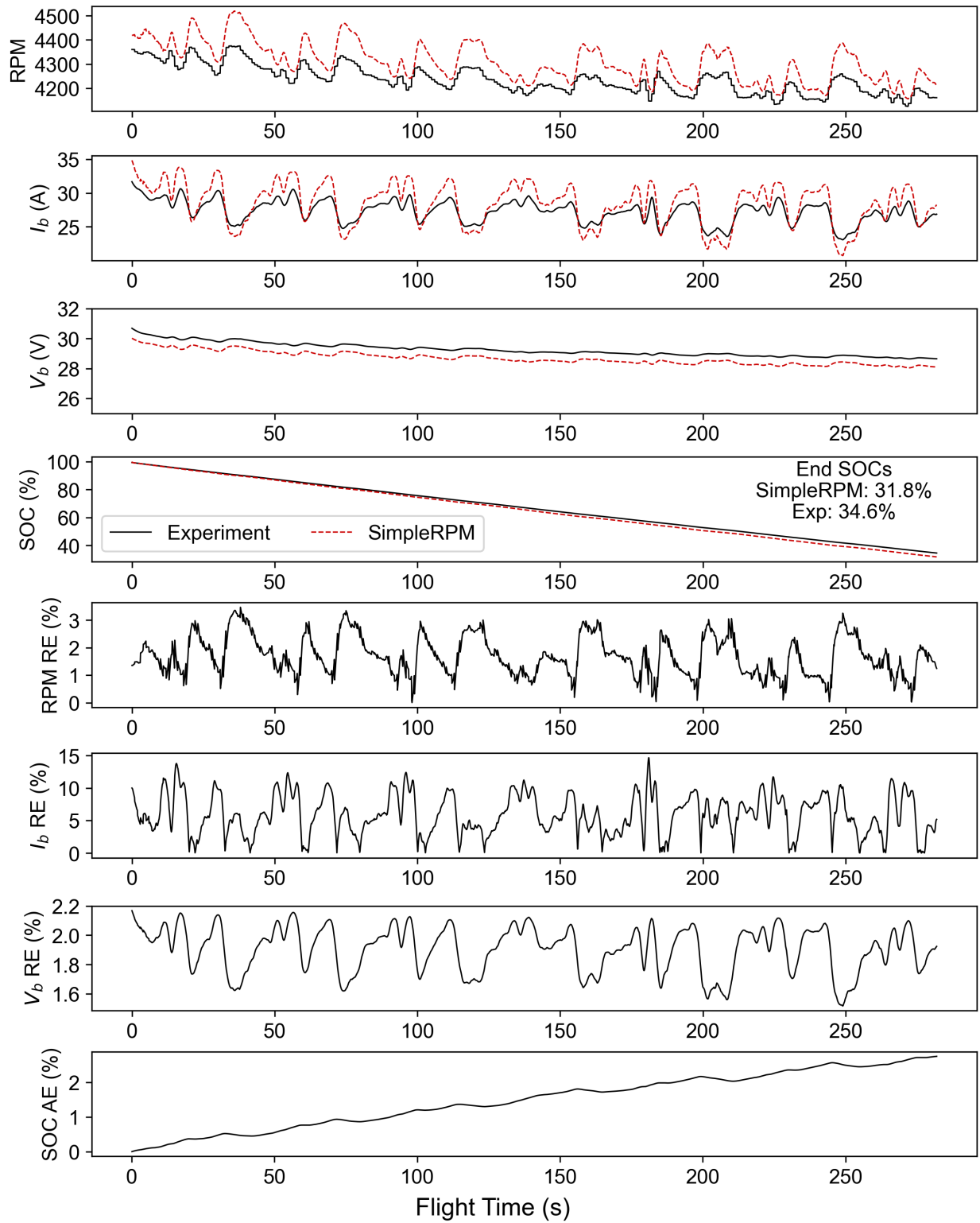


Fig. 22 Flight 3 data vs SimpleRPM with relative errors (RE) and absolute errors (AE).

Table 4 Absolute error (AE) metrics for state of charge (SOC) in %.

Comparison	Max AE	Mean AE	Median AE
Flight 1	5.8840	3.2716	3.5234
Flight 2	2.2594	1.1920	1.1778
Flight 3	1.6036	0.9058	0.9986

IV. Case Studies

In this section, a demonstration of the SimpleRPM model and the UAVDEX toolbox applied to design and analysis of RPI DBF's 2026 aircraft is presented. For the 2025-26 AIAA DBF competition held in Wichita, KS, the second flight mission (M2) was scored based on the number of passengers (n_{pax} rubber ducks), the number of cargo pieces (n_{cargo} hockey pucks), and the number of laps flown in a five minute time limit (n_{laps}). The third flight mission (M3) was a banner towing mission. The aircraft was required to deploy a banner in-flight and to fly as many laps as possible within the five minute time limit. Scoring for M3 was calculated by,

$$M3 = 2 + \frac{n_{\text{laps}} \cdot L_{\text{banner}}}{\max M3} \quad (7)$$

where $\max M3$ is the maximum $n_{\text{laps}} \cdot L_{\text{banner}}$ value achieved by any team, and L_{banner} is the banner length in inches. The DBF flight course is organized as shown in Figure 23.

A. Conceptual Design Optimization with the SimpleRPM Model

For the 2026 DBF competition, the first author developed a optimization to preliminarily select payloads, determine a generic propulsion system, size the wing and fuselage, and generate initial performance predictions. This work is detailed in the RPI DBF's design report*. A key enabler for the optimization was the SimpleRPM model, applied within a simultaneous analysis and design (SAND) framework, which used the mixed-integer nonlinear problem capable APOPT algorithm [29] via the Python GEKKO package [30]. The model determined T , I_b , and η_{drive} for cruise and level turn segments for M2 and M3. That, in turn, allowed for prediction of cruise velocity and n_{laps} . Figure 24 visualizes the optimization results for a range of fixed n_{pax} . These results prompted the RPI DBF team to select the higher scoring minimum n_{pax} design. Out of the top 7 teams[†], 6 of them selected this concept, thereby validating the result of this optimization.

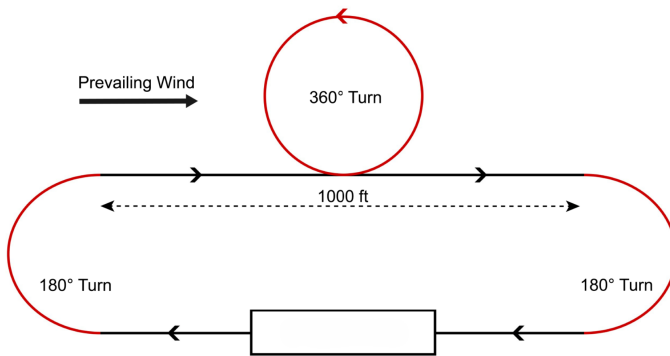


Fig. 23 Design/Build/Fly competition flight course.

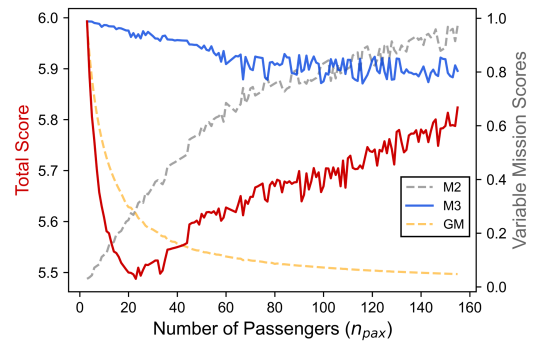


Fig. 24 Optimized score variation with n_{pax} .

This optimization converged on a propulsion system with an 8S 3300 mAh battery, an APC 22x12E propeller, and one motor with a K_V of 170 and maximum power of 1100 W. One critical constraint applied was that the minimum airspeed during flight must exceed 30 mph to ensure adequate ground speed in Wichita gusts.

* See Section 4 of <https://tinyurl.com/57jbsb92>

[†] See <https://aiaa.org/wp-content/uploads/2026/04/2026-DBF-Competition-Summary.pdf>

B. Component Specified Analyses

Following from the aeropropulsive optimization, a group of candidate motors close to the desired K_V and maximum power were gathered from an internal database. These motors and a group of APC propellers were investigated for cruise and turn conditions in UAVDEX using the RuntimePlot and ContourPlot functions. Examples of these analyses are shown in Figures 25-27 for the selected propulsion drive consisting of a SunnySky V8110-170 motor, Gaoneng 8S 3300 mAh battery, and APC 22x12E propeller.

The RuntimePlot function takes in a constant altitude or air density and plots the time (s) versus U_∞ (mph) with lines at different δT (%). This is achieved by solving the SimpleRPM model along a range of velocities for the altitude/density, throttle, and final SOC (typically 20% for LiPo batteries). Given a drag coefficient (C_D) and planform wing area (S_w), a contour of $T = D$ was plotted. As shown in Figure 25, the propulsion system was designed to run at full throttle to maximize efficiency (see Figure 26 for the relationship between velocity, throttle setting, and efficiency). For the steady-state condition using the cruise C_D plus the banner drag, a five minute (300 s) runtime was selected to provide some margin for landing (orange dashed line in Figure 25).

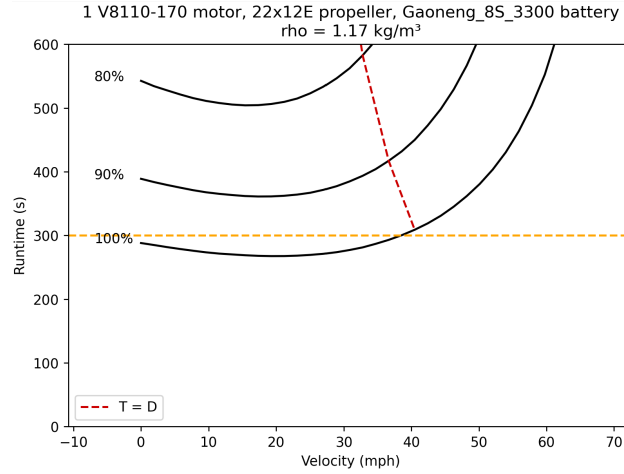


Fig. 25 UAVDEX RuntimePlot for the RPI DBF 2026 propulsion drive. Each black line corresponds to a fixed throttle labeled by the % on the left.

The ContourPlot function takes in two constant values and two ranges of the U_∞ , δT , h , and t inputs, then plots a propulsion quantity such as T , η_{drive} , or I_b as a filled contour over the provided ranges. It does so by solving the SimpleRPM model at each of the fixed four inputs. Constraints based on the motor's maximum continuous power (in W), the battery's continuous current (in A), and the propeller tip speed are automatically plotted. Additionally, datatips are available in the interactive Python plots.

For the RPI DBF 2026 propulsion setup, Figure 26 plots η_{drive} as a function of velocity and throttle, while Figure 27 plots η_{drive} as a function of velocity and runtime. As indicated earlier with the RuntimePlot, at full throttle the end runtime is slightly over five minutes; this runtime is represented as the boundary between the color contours and the white region. Critically, Figure 27 also shows that the velocity for the $T = D$ line never drops below 30 mph, so the propulsion system meets the ground speed constraint set in the optimization. Moreover, it indicates a high propulsive efficiency of 62% across the flight mission.

The continuous power limits of BLDC motors can be exceeded for under 30s, which is useful for competition aircraft. Using Figure 27, the 1425 W constraint (red curve) near the left-hand side of the plot shows that the propulsion system exceeds the stated motor power limit between 0-40 mph in the first 10s of flight. With this information, RPI DBF concluded that since the power limit of the motor was only exceeded for a brief period of time, this setup was still viable. This conclusion was proven through successful flight tests of the propulsion system. If relying on an RC calculator, a designer would only know that the power limit was exceeded and might have pivoted to a suboptimal design.

These interconnected analyses of efficiency, sustained velocity, and runtime over the full flight envelope are not possible to determine easily with any other open-source tool. Other tools would require significant iteration, while in UAVDEX, a user simply selects some components then calls three functions (see the Appendix for the exact code used).

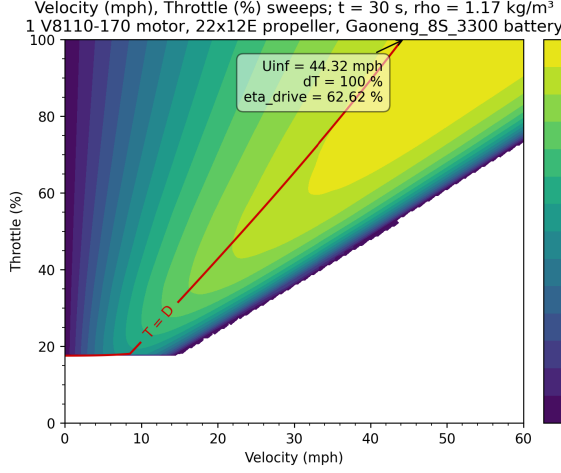


Fig. 26 UAVDEX η_{drive} plot against $U_{\infty}, \delta T$

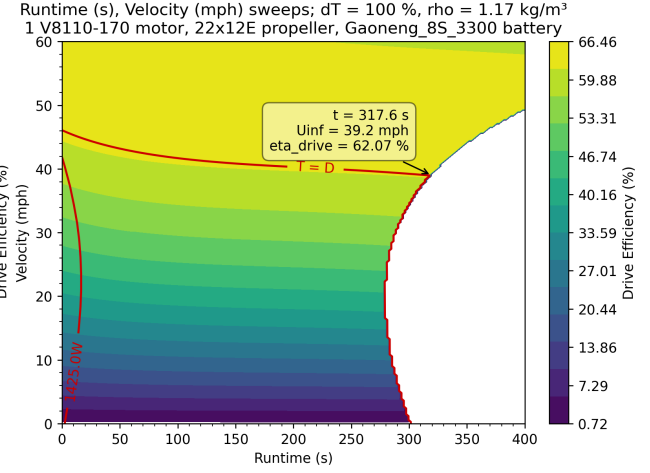


Fig. 27 UAVDEX η_{drive} plot against U_{∞}, t

V. Conclusion

We presented an accessible new design tool for electric fixed-wing aircraft alongside validation of a proposed propulsion model and case-study applications. It was found that the SimpleRPM model produced runtime estimates within 15% of the experimental results, but consistently underpredicted thrust by ~25% primarily due to error from APC BEMT C_T data. With experimentally provided C_T data, the thrust prediction error decreased to 11%. Despite these flaws, the SimpleRPM model proved to be as accurate as eCalc while providing a simple open-source implementation that can be easily integrated in Python-based MDAO codes.

This work is far from state-of-the-art among all codes intended for MDAO of small electric vehicles, but it fills an important niche as an accessible, open-source tool for student design teams and hobbyists. With the spreadsheet-based component databases, the tool has the potential to remain relevant and adaptable for a long while. It is the hope of the authors that the [UAVDEX code](#) will find use among DBF teams and increase the technical rigor and understanding by which teams select their propulsion systems.

Acknowledgments

The authors greatly appreciate the support of the RPI DBF team. Without their passion and dedication, this paper would not exist. Specifically, we acknowledge James Zeolla and Rudra Pal for their work in creating custom flight instrumentation and conducting experiments to gather propulsion data.

Appendix

<https://github.com/sammy183/UAVDEX>

```
import uavdex as ud
import numpy as np

# initialize analysis and define components
design = ud.PointDesign()
design.Motor('V8110-170', nmot = 1)
design.Battery('Gaoneng_8S_3300', discharge = 85)
design.Prop('22x12E')

# set drag coefficient (CD) and reference wing area (Sw) in ft2
CD = 0.22
Sw = 8.781

# create runtime plot with target runtime of 300s
# and maximum shown runtime of 600s
design.RuntimePlot(rho_kgm3 = 1.17,
                  CD = CD, Sw_ft2 = Sw,
                  t_target = 300,
                  t_limit = 600,
                  verbose = False)

# set number of points per input sweep
n = 200

# create efficiency contourplot for Uinf & dT, set t = 30s, rho = 1.17 kg/m3
design.ContourPlot(propQ = ['eta_drive'],
                  Uinf_mph = np.linspace(0, 60, n),
                  dT = np.linspace(0, 100, n),
                  t_s = 30,
                  rho_kgm3 = 1.17,
                  CD = CD,
                  Sw_ft2 = Sw
                  )

# create efficiency contourplot for Uinf & t, set dT = 100%, rho = 1.17 kg/m3
design.ContourPlot(propQ = ['eta_drive'],
                  Uinf_mph = np.linspace(0, 60, n),
                  dT = 100,
                  t_s = np.linspace(0, 400, n),
                  rho_kgm3 = 1.17,
                  CD = CD,
                  Sw_ft2 = Sw
                  )
```

Fig. 28 Code for Figures 25-27

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